BHAVISHYA VUDATHA

AI/ML Engineer

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SUMMARY

AI/ML Engineer with 4+ years of experience designing and deploying scalable **machine learning**, **Generative AI**, and **MLOps solutions** across cloud environments. Expertise in **Large Language Models (LLMs)**, **Retrieval-Augmented Generation (RAG)**, **NLP**, **predictive modeling** and real-time inference pipelines using **Python**, **TensorFlow**, **PyTorch,** and **AWS services**. Proven track record of improving model accuracy, accelerating deployment cycles, and enhancing production reliability through automated **ML workflows**, **monitoring**, and **cloud-native architectures**. Skilled at transforming complex data into high-impact **AI solutions** that drive operational efficiency, user adoption, and measurable business outcomes.

SKILLS

Machine Learning & AI: Supervised Learning, Predictive Modeling, Feature Engineering, Classification, Regression, Clustering, Recommendation Systems, Time Series Forecasting, Anomaly Detection, Model Evaluation, Deep Learning, Statistical Modeling, NLP

Generative AI & NLP: Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Text Embeddings, Semantic Search, NLP Pipelines, Transformer Architectures, Fine-Tuning, AI Chatbots, LangChain, LangGraph, Hugging Face, AI Agents, Claude

MLOps & Model Deployment: MLflow, Model Monitoring, Drift Detection, Experiment Tracking, CI/CD for ML Pipelines, Real-Time Inference Pipelines, Automated Model Retraining, Production Model Deployment

Programming & Software Development: Python, SQL, Java, JavaScript, TypeScript, C++, Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, XGBoost, React, Node.js, REST APIs, Git, Object-Oriented Programming, System Design, Data Structures & Algorithms

Data Engineering & Processing: ETL Pipelines, Apache Spark, Hadoop, MapReduce, Apache Airflow, Apache Kafka, AWS Glue, Amazon Kinesis, Data Pipeline Automation, Feature Data Processing

Databases & Data Warehousing: MySQL, PostgreSQL, MongoDB, Oracle, Snowflake, Amazon Redshift

Cloud & AI Infrastructure: AWS SageMaker, AWS Bedrock, AWS Lambda, EC2, S3, API Gateway, Serverless AI Architecture, Cloud-Based Model Serving, Azure (ADLS, ADF, Synapse, Blob Storage), GCP

Monitoring, Reporting & Visualization: Prometheus, Grafana, AWS CloudWatch, AWS QuickSight, Tableau, Power BI, KPI Reporting, Data Visualization

EDUCATION

Master of Science in Computer Science — **Dec 2025**
University of Colorado Denver, USA

B. Tech in Computer Science — **May 2019**
Jawaharlal Nehru Technological University

EXPERIENCE

ServiceNow, CA | AI/ML Engineer — Jul 2025 – Current

- Architected a **Retrieval-Augmented Generation (RAG)** platform leveraging **LLMs**, **Hugging Face Transformers** and vector embeddings, improving enterprise knowledge retrieval accuracy by 42% and reducing response latency by 35%.
- Engineered real-time inference pipelines on **AWS SageMaker**, **Lambda,** and **API Gateway**, supporting 1M+ monthly **AI requests** while maintaining 99.9% service availability.
- Optimized predictive **machine learning** models through advanced feature engineering and **hyperparameter tuning**, increasing classification performance by 28% and reducing false positives by 31%.
- Automated end-to-end **MLOps workflows** using **MLflow**, **CI/CD pipelines,** and **experiment tracking**, decreasing model deployment cycles by 65% and accelerating releases across multiple AI products.
- Implemented model monitoring, drift detection, and automated retraining frameworks, improving production model stability by 40% and reducing performance degradation incidents by 55%.
- Spearheaded development of AI-powered chatbot solutions utilizing transformer architectures and semantic search capabilities, increasing user self-service adoption by 48% and reducing support workload by 33%.

Orion Technolab, India | ML Engineer — Jan 2020 – Feb 2023

- Developed **scalable machine learning models** for classification, regression, and recommendation systems, improving prediction accuracy by 24% and increasing customer engagement by 19%.
- Built automated **ETL** and feature processing pipelines using **Python**, **SQL**, **AWS Glue**, and **Amazon Kinesis**, reducing data preparation time by 60% and supporting multi-terabyte datasets.
- Designed anomaly detection frameworks for operational analytics, identifying critical system deviations 45% faster and reducing incident resolution times by 30%.
- Integrated **NLP pipelines** and transformer-based language models for text analytics initiatives, improving document processing efficiency by 52% and enhancing information extraction accuracy by 27%.

- Leveraged **TensorFlow**, **PyTorch**, **XGBoost**, and **Scikit-learn** to create production-grade forecasting solutions, reducing forecasting errors by 22% across business-critical datasets.
- Established cloud-native model serving infrastructure on **AWS EC2** and **S3**, enabling horizontal scalability and lowering infrastructure costs by 26% through resource optimization.
- Generated executive KPI dashboards using **Grafana**, **AWS QuickSight**, **Prometheus**, and **CloudWatch**, increasing reporting visibility by 70% and accelerating data-driven decision-making across stakeholders.

ACADEMIC PROJECTS

YouTube Data Analytics Pipeline | S3, Lambda, Glue Studio, Athena, Step functions, PySpark, IAM, crawler

- Built an automated **AWS ETL pipeline** to analyze factors influencing YouTube video views by ingesting region-wise data into an S3
- Processed and transformed raw data using **AWS Lambda (Python)** and **AWS Glue (PySpark)**, converting datasets into optimized Parquet format for efficient querying
- Created and managed metadata using **AWS Glue Data Catalog and crawlers**, enabling schema discovery and query access via Athena
- Orchestrated end-to-end workflows using **AWS Step Functions**, enabling automated data processing triggered by new data ingestion
- Performed data transformations and joins on **JSON** and **CSV datasets** using **Glue ETL jobs**, generating analytics-ready datasets for visualization in Tableau/Power BI

Real-Time Telecom Analytics Pipeline | Databricks, PySpark, Delta Lake, AWS S3, Airflow

- Built a real-time telecom analytics pipeline processing 120K+ daily records to analyze churn and network insights, using **Databricks Autoloader** and **Medallion Architecture** (Bronze–Silver–Gold) on **AWS S3** and **Unity Catalog**
- Orchestrated ingestion of 800+ daily **CSV/JSON** files using **Apache Airflow (Docker)** and built **DLT pipelines** to transform and join data in the gold layer into reliable, analytics-ready datasets
- Reduced end-to-end pipeline latency to 5 minutes by implementing event-driven orchestration, using **Databricks Jobs API**, file-based triggers, and idempotent workflow logic

AI Email Processing Agent | Python, Gemini/Claude LLM, PostgreSQL, AWS

- Developed a job application tracking agent using **Python** and **Claude LLM** to ingest emails via **Gmail API**, classify job-related emails using prompt engineering, and extract structured data (company, role, status) into **PostgreSQL** with zero manual intervention
- Designed an **LLM-based email classification pipeline** categorizing emails across 7 job statuses with semantic rejection detection
- Building a conversational **AI chatbot** using **Claude LLM** connected to **PostgreSQL**, enabling natural language queries on job application data (e.g., "which companies rejected me?" or "how many applications this month?") without writing **SQL**.
- Productionizing on **AWS (Lambda, RDS PostgreSQL, S3, EventBridge, CloudWatch)**, containerizing with **Docker** Compose, and building a Streamlit dashboard with real-time analytics.

Multimodal RAG Pipeline for Research Papers

- Built a multimodal retrieval system processing 239 arXiv papers using **Marker**, **CLIP** and **BGE embeddings** indexed in **Qdrant**. Implemented **LangGraph 8-node pipeline** with cross- encoder reranking, HyDE optimization, and dual-layer memory. Used Claude Haiku for grounded responses and evaluated with RAGAS, achieving a 0.75 faithfulness score. Deployed via Streamlit

Cloud-Based AI Email Processing Agent

- Built an AI job tracking agent integrating Gmail API with Claude Haiku for classification across 7 application statuses and ATS parsing. Stored structured results in **PostgreSQL** via **FastAPI** with deduplication, rate limiting, and retry mechanisms for reliable processing

SpaceX Falcon 9 Landing Prediction –

- Developed a **machine learning pipeline** leveraging supervised models (Random Forest, Logistic Regression, SVM, Decision Tree) to predict Falcon 9 first stage landing success using historical **SpaceX API data**. Performed data wrangling and EDA using **Matplotlib** and **Seaborn**. Fine-tuned models with **GridSearchCV**, outperforming baseline classifiers

CERTIFICATIONS

- **Machine Learning Specialization, Andrew Ng**
- **Databricks Certified Data Engineer Associate**